SET C

Q. No	Question
1	int *arr = malloc(5 * sizeof(int)); Which of the following correctly reallocates arr to double its size? A. realloc(arr, 10 * sizeof(int)); B. arr = realloc(arr, 10 * sizeof(int)); C. realloc(10 * sizeof(int), arr); D. arr = malloc(10 * sizeof(int)); memcpy(arr, arr, 5*sizeof(int));

2	int x = 10;
	int *p = &x;
	
	Which of the following correctly prints the address of variable x?
	
	A. printf("%d", p);
	B. printf("%p", p);
	C. printf("%p", x);
	D. printf("%d", &x);

3	What is the time complexity for the following code snippet
	for (i = 1; i < n; i = i * 2) {
	printf("%d ", i);
	}
	A. O(n)
	B. O(log n)
	C. O(n log n)
	D. O(n²)

4	int *p;
	p = (int *)calloc(5, sizeof(int));
	printf("%d", p[3]);
	
	What will be the output of the given code?
	
	A. Garbage value
	B. Compilation error
	C. 0
	D. Segmentation fault

5	The total number of scalar multiplications required to multiply A (n×k) and B (k×m): A. O(n + m + k) B. O(nkm) C. O(nk^2m) D. O(nm)
6	Consider the following code, int main()

	{
	int a = 10, b = 20, c = 30;
	int *arr[3];
	arr[0] = &a;
	arr[1] = &b;
	arr[2] = &c;
	---------- 	
	return 0;
	}
	
	Which of the following option correctly print the value 20?
	
	A. printf("%d", *arr[1]);
	B. printf("%d", *(arr + 1));
	C. printf("%d", arr[1]);
	D. printf("%d", **(&arr[1]));

7	Let f(n)=3n^2+7n+4 Which of the following is a tight upper bound for f(n)? A. O(n^3) B. O(nlogn) C. O(n^2) D. O(1)
8	If an algorithm runs in time $T(n) = 3n^2 + 2n + 1$, which of the following is true? A. $T(n) = O(n^2)$ B. $T(n) = \Omega(n^2)$ C. $T(n) = \Theta(n^2)$ D. $T(n) = \Omega(n)$

9	struct Inner {
	int a;
	};
	struct Outer {
	struct Inner in;
	int b;
	};
	Which among the listed are true?
	A. Outer contains a complete Inner struct
	B. Memory for Inner is allocated inside Outer
	C. Accessing Outer.a is valid
	D. Accessing Inner in.a is invalid

10	struct Student {
	int marks;
	};
	void update(struct Student *s) {
	s->marks += 10;
	}
	int main() {
	struct Student s = {50};

	update(&s);
	printf("%d", s->marks);
	}
	Output of the program?
	A. 50
	B. 60
	C. Garbage value
	D. Compilation error

SET D

Q. No	Question
1	Given a structure in C, how would you dynamically allocate memory for an array of 10 such structures? A. struct myStruct arr[10]; B. struct myStruct* arr = (struct myStruct*)malloc(10 * sizeof(struct myStruct)); C. struct myStruct* arr = malloc(10); D. struct myStruct arr = malloc(10 * sizeof(struct myStruct));
2	Which of the following is true about an Abstract Data Type (ADT)? A. It is a concrete implementation of data structure. B. It specifies the logical behavior of a data structure without specifying the implementation. C. It can only be implemented using arrays. D. It does not support any operations.

3	What is the output of the following code?
	void main()
	{int *p = NULL;
	p = realloc(p, sizeof(int) * 2);
	printf("%d %d", p[0], p[1]);
	p[0] = 1;
	p[1] = 2;}
	
	A. 1 2
	B. 0 0
	C. Segmentation fault
	D. Garbage

4	struct Node {
	int data;
	struct Node next;
	struct Node *next;
	};
	Based on the code above, which of the following statements are correct?
	
	A. The line struct Node next; is valid in C and creates a new Node inside Node.
	B. The line struct Node next; will cause a compilation error due to incomplete type.
	C. The line struct Node *next; is valid and defines a pointer to another Node.
	D. A structure can contain a pointer to itself but not a direct instance of itself.